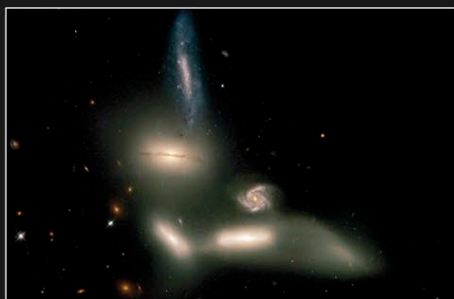


GALAXY COLLISIONS

Relative to their size, galaxies are quite closely packed together—although they are separated by distances of hundreds of thousands of light-years, galaxies themselves are typically tens of thousands of light-years across. Furthermore, the enormous gravity exerted by large galaxies and their tendency to form within large-scale clusters allow them to influence and attract one another. As a result, collisions and close encounters between galaxies are comparatively common. In 1966, US astronomer Halton Arp compiled the first catalog of galaxies that did not fit neatly into the common categories of spiral, irregular, and elliptical. With the benefit of more recent observations, it now seems that most of Arp's unusual galaxies were the result of past collisions and interactions between galaxies. Even some apparently normal galaxies are now thought to have interacted with other galaxies in the past, and it is also clear that many large galaxies are “cannibals,” tearing apart and ultimately absorbing smaller galaxies that stray too close. However, during intergalactic collisions, individual stars rarely collide, and it may take several billion years before the mutual gravity of colliding galaxies finally pulls together most of their material into a single combined cloud of stars.



SEYFERT'S SEXTET

Despite its name, this group contains only five galaxies—the bright patch on the right is an unwinding spiral arm. Only four of the galaxies are at the same distance from Earth, about 190 million light-years away; the face-on spiral galaxy is about five times that distance. The four nearest galaxies are being distorted by gravitational forces between them.

THE SPLINTER GALAXY

Also known as the Knife Edge Galaxy or NGC 5907, this galaxy is an edge-on spiral that lies about 40 million light-years from Earth in the constellation Draco. It is surrounded by extraordinary looping trails of faint stars, nicknamed the “Ghost Stream,” which are thought to be the remnants of a smaller galaxy that has now been consumed by NGC 5907.



COLLISIONS AND EVOLUTION

The process of collision is now thought to play a key role in transforming galaxies from one type to another. In the early stages of a collision, stars that may have had relatively orderly orbits are pushed into highly elongated and tilted paths, and powerful shock waves passing through interstellar gas and dust generate tremendous bursts of new star formation. In the longer term, the remaining gas may become energized to such a degree that it can escape the galaxy's gravity altogether, depriving it of the means to continue star formation. In this way, spiral and irregular galaxies can be transformed into ellipticals surrounded by clouds of hot gas, as seen in the central regions of many galaxy clusters. However, it has also been theorized that this process can be reversed, at least in the relatively short term. According to this theory, cold intergalactic gas is constantly drawn in by the galaxy's gravity and can ultimately form a flattened disk in which star formation can begin again and spiral arms reform. If this theory is correct, then the rare lenticular galaxies mark an intermediate phase between elliptical and spiral galaxies. However, over time, merging spiral galaxies will form larger and larger ellipticals, while dwindling reserves of cold intergalactic gas will slow the regeneration of spiral galaxies.

COLLIDING GALAXIES

Situated about 450 million light-years from Earth in the constellation Hercules, two spiral galaxies (NGC 6050 and IC 1179, also collectively known as ARP 272) are colliding. Tidal forces in both galaxies are triggering enormous waves of star formation, manifested in the bright clusters around their spiral arms.



SPIRALS IN COLLISION

The galaxy NGC 520 (also known as Arp 157) in the constellation Pisces is believed to be a pair of colliding spirals seen edge-on. The collision began about 300 million years ago, and the galaxies are now in the middle stages of merging: their disks have come together, but their nuclei have not yet merged.